

Quinlan Lewis

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--- EDUCATION ---

B.S. in Aerospace Engineering Sciences

Spring 2022

University of Colorado Boulder, Boulder, CO

- GPA: 3.43/4.0
- Minors: Computer Science & Space

--- PROFESSIONAL EXPERIENCE ---

Software Engineer II, Raytheon Intelligence & Space, Aurora, CO

2023 - Present

- Led the migration of an internal **Angular**-based UI component to **React** for cross-application usage; implemented comprehensive unit tests and built a **Storybook** app for documentation and visual testing.
- Developed and integrated multiple **RESTful APIs** and added **WebSocket** support for select backend services to enable real-time data flow.
- Improved code coverage across multiple projects in a **Java Spring** backend service integrated with **PostgreSQL**, contributing to system reliability and test quality.
- Deployed and maintained a complex multi-component system on a **Kubernetes** cluster using **Helm** charts and **Docker** images; configured ingress controllers and rules to support scalable access.
- Ensured **CI/CD** pipeline integrity by writing unit tests and managing **Jenkins** jobs, keeping builds stable and test suites up-to-date.

Software Engineer I, Raytheon Intelligence & Space, Aurora, CO

2022 - 2023

- Integrated a COTS mapping **API** into a thin-client web application supporting **satellite system management** as a subsystem of a larger governing program; developed with **JavaScript**, **TypeScript**, and **CSS** in an **Angular** and **React** environment.
- Enhanced and customized reusable software to better align with program requirements, including the development of **new features** that improved **user flexibility and customization** within the web application.
- Deployed services and **Docker** images to **Kubernetes** clusters using **Helm charts**, streamlining the application's infrastructure delivery.
- Participated in the **Agile software development lifecycle** using **Jira** and other **Atlassian tools**, contributing to an **iterative, cross-functional team environment** focused on flexibility and collaboration.
- Conducted trade studies to identify the most suitable front-end framework and supporting technologies for program needs; compared **Vue.js**, **Angular**, and **React** along with other **open-source and COTS solutions**.

Undergraduate Research Assistant, BioServe Space Technologies, Boulder, CO

2021 - 2022

- Developed and designed a Virtual Reality simulation of The Lunar Rover using Unity programming in **C#** to test stochastic resonance as a use in spaceflight countermeasures
- Expanded project scope to include development of a procedural generated terrain maps to help further research by adding an aspect of random noise into testing; Maps were created in **Unity** with **C#** for implementing white noise
- Ending publication - <https://www.frontiersin.org/articles/10.3389/fnins.2023.1180314/full>

--- PROJECTS ---

Software Systems Integration Lead, Senior Capstone Project, Boulder, CO

2021 - 2022

- Lead implementation of a software controller for an autonomous high endurance UAV system; goal of project was to increase endurance of an autonomous UAV leveraging only software
- Designed, tested, and tuned Linear Quadratic Tracker controller through **MATLAB** and converted to **C++** for integration

--- TECHNICAL SKILLS ---

- Programming Languages: C | C++ | C# | JavaScript | TypeScript | CSS | HTML | Python | Java | Yaml | Bash
- Other Program Experience: MATLAB | Microsoft Office Suite(Word, PowerPoint, Excel) | Atlassian Suite(Jira, Bitbucket, Confluence) | Systems Tool Kit(STK) | Git| Unity | Gradle | Docker | Kubernetes | Helm | Vue.js | React | Node.js | Angular | Markdown